Formal Lab Report - Identification of Unknown B

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Introduction

The goal of this lab report is to analyze Mass Spectroscopy, nuclear Magnetic Resonance, and Infrared Spectroscopy data to identify the assigned unknown compound. Mass Spectroscopy, Nuclear Magnetic Resonance, and Infrared Spectroscopy are three methods used in organic chemistry to anay;lize and identify unknown chemical compounds or to distinguish compounds that have the same chemical formula.

Body

1). Mass Spectroscopy (MS) is the study of interactions between matter and energy sources and is often used to analyze molecules and particles according to mass off the chemical compound.The simplest setup of a mass spectroscopy can be referred to a sample that is introduced in the gas phase and these molecules are bombarded by an electron beam and the molecules get transformed into charged particles. When a magnetic field is applied, these charged particles will be deflected according to their mass and focused on a detector. When a molecule is ionized by an e-beam, a radical cation is formed, this is the molecular ion. As a result, the molecular ion will fracture into small pieces due to the inherent instability of charged particles. The molecular ion and all the fragment ions are analyzed according to their mass/charge ratio, which will yield the structural information about the analyte. All the cations that are deflected are detected and used to create a mass spectrum. Then the mass spectrum shows the relative abundance of each cation that was detected. The tallest peak on the mass spectrum is deemed the base peak, and every other peak is then described relative to the height of the base peak. When analyzing the mass spectrum, firstly, we need to look at (M)+ because it tells us the molecular weight of the molecule. The (M+1) + peak and the (M+2) + peaks help to determine the unknown compound. And if an organic chemical compound has a chlorine atom,

the (M+2) + peak will be approximately 1/3rd the height of the molecular ion peak, meanwhile if there is a presence of a bromine atom in the compound, the (M+2) + peak will be about the same height as the molecular ion peak.

2) Infrared Spectroscopy (IR) is studying the infrared light interacting with molecules. This method is used for wavelengths in the range of 2.5x10-4 cm to 2.5x10-3 cm and for IR spectra the position of the signals is expressed in wavenumbers and the useful IR region is from 4,000 to 400 cm-1. And when the organic molecules absorb radiant energy in the IR region, specific bonds within the molecule become excited and vibrate. If the frequency of the incident radiation matches the specific vibrational frequency of the bond, the bond is excited to higher vibrational state and the amplitude of the vibration increases as well. Molecules with different structures do not absorb the same energies of infrared radiation and therefore they give different patterns of absorption. IR spectroscopy is preceded by irradiating a sample with IR light and using IR spectrometer, when the light passes through the sample, the light is sent to the detector that measures frequencies that were not absorbed by the unknown organic compound. Then the data gets converted to an IR plot, which shows the frequencies that were absorbed. The most common spectrometer used by chemists is Fourier Transform Spectrometer (FT-IR), it simultaneously irradiates a sample with the frequencies and uses a mathematical operation to obtain the whole spectrum at one time

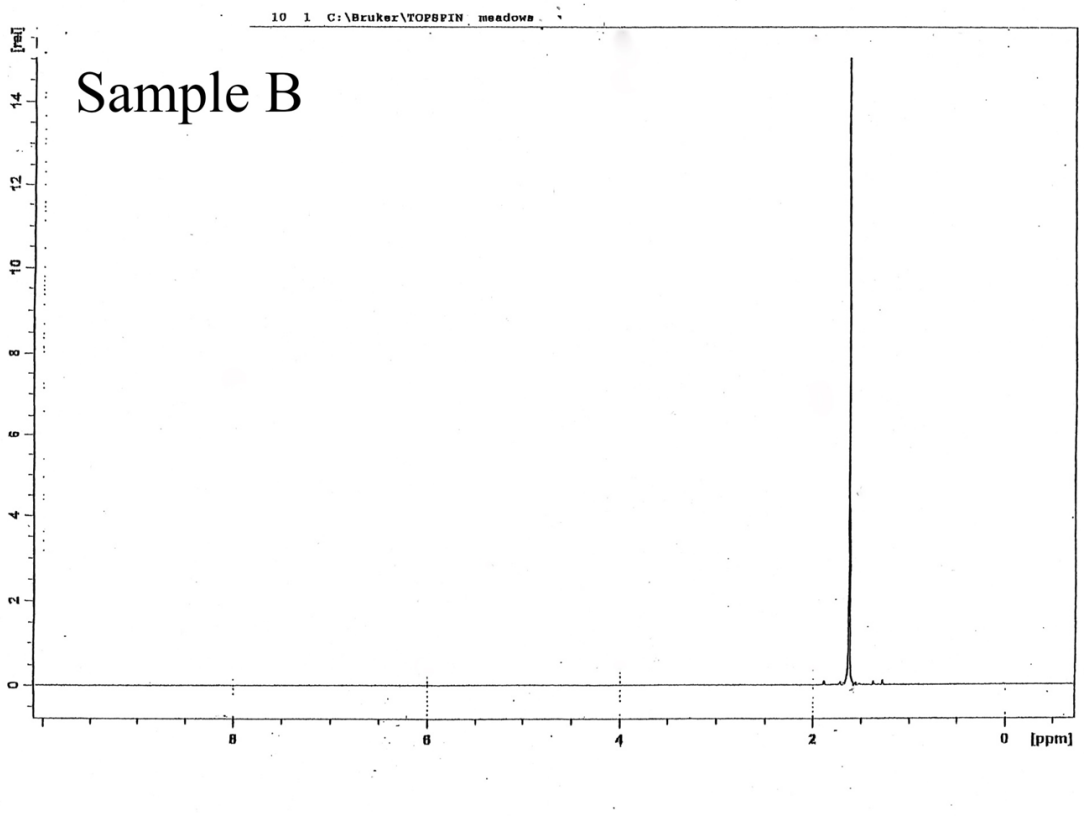
3) Nuclear Magnetic Resonance (NMR)

connectivity of atoms. NMR spectroscopy can be used to determine the exact location of functional groups and hydrogens, give valuable information about the carbon skeleton and about the molecule. NMR spectroscopy might apply for any nucleus that has an odd mass number and/or odd atomic number. The most common type of NMR is 1H (proton) NMR, otherwise 13C

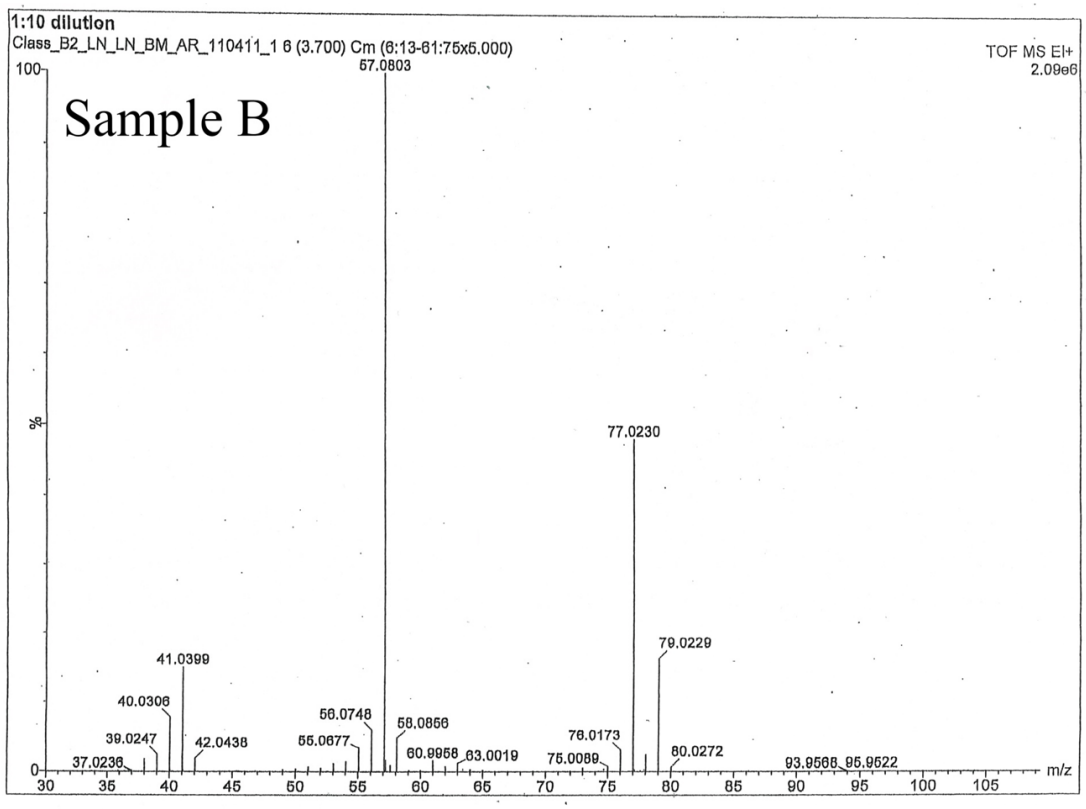
NMR is also a common use. For 1H NMR, users would look for the most abundant isotope of hydrogen, while in the case of 13C NMR the 13C isotope only represents roughly 1% of the carbon atoms.1 The first object that is needed to look at when analyzing NMR is the number of signals present on the spectrum, because each signal indicates a kind of proton that is in a unique electronic environment2. Secondly, we would need to care about the signal's location, which is also called "chemical shift" and defined relative to the frequency absorption of a reference compound, tetramethylsilane (TMS). Another thing to concern is the area under each signal, because it can give the number of protons giving rise to that signal. The last thing we would want to look at is the number of peaks at a particular signal, which is called multiplicity. A signal's multiplicity is a result of the magnetic effects of neighboring protons and therefore indicates the number of neighboring protons. If a neighboring proton is in a different environment, the result would couple. And if all the protons are chemically equivalent, then the (n+1) rule can be applied. (n+1) rule claims that the number of peaks at a signal equal to (n+1), when n is the number of neighboring protons.

Data

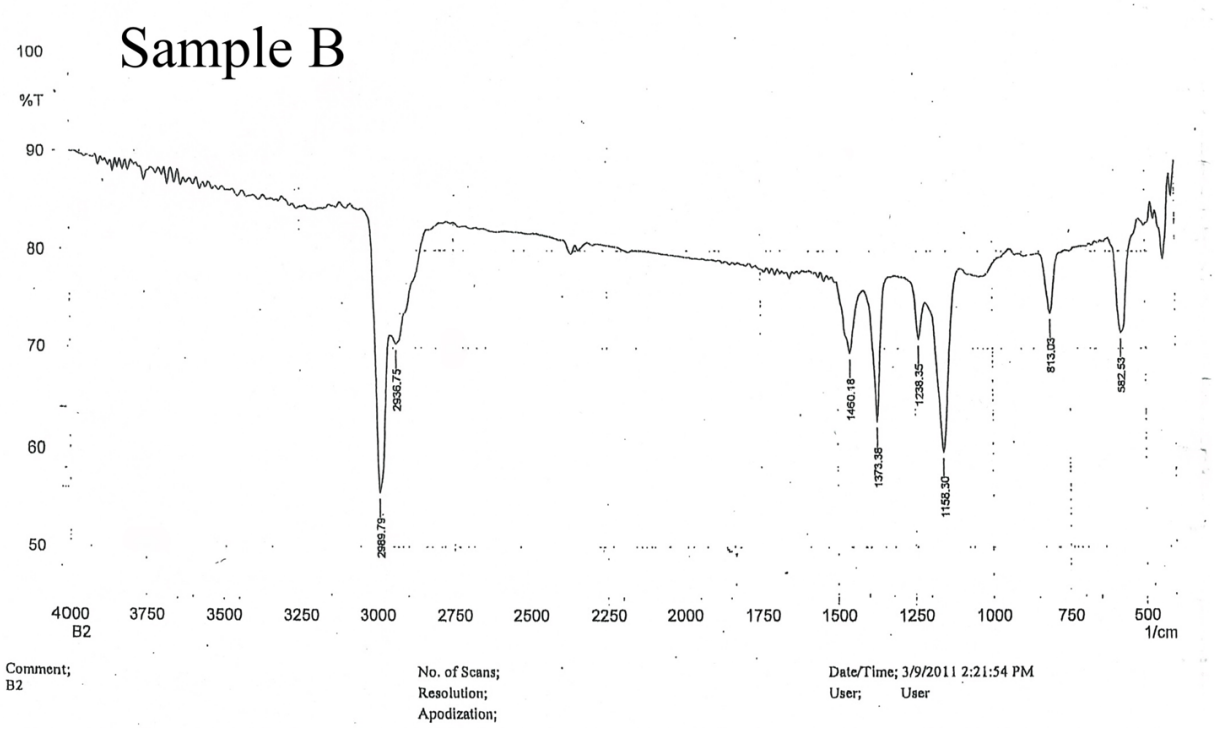
NMR sample B



GCMS sample B



FTIR Sample B



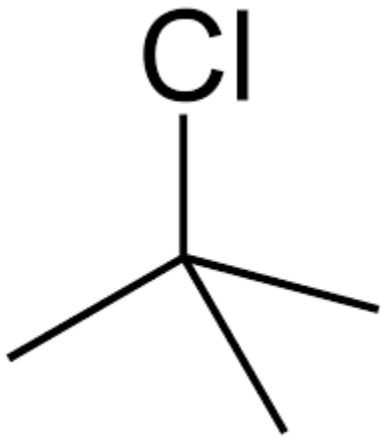
Analysis

For NMR, the graph shows a single peak at about 1.6 ppm and the primary protons peaks at 1.11.2 ppm, it can be concluded that there is a presence of chloride atom in the compound, and it causes a chemical downfield shift. The presence of a single peak tells that there is only one type of proton present in the compound. Furthermore, the signal is tall, meaning that there are multiple protons that are giving rise to the signal. For FTIR, there is no activity above 3000 cm -1, which means there is no alcohol group in the compound. The peaks at 2989.79 and 2936.75 indicate that there are C-H bonds in the compound. Also, there is no peak at 1600-1700 cm-1, which means there are no double or triple bonds. There are some signals between 1500-100 cm-1, which identify the methyl group. The peaks at 813.03 and 582.83 indicate that there is a C-Cl bond. For GCMS, the base peak is at 57 m/z, which means the most stable and possible cation chemical formula is C4H9. (M+1) + peak is at 77 and (M+2) + peak is at 79, in which

(M+2)+ is about 1/3 the height of (M+1)+, which indicates the chlorine atom. Moreover, the molecular ion peak is found at 93, which indicates the chemical formula of tert-Butyl Chloride.

Conclusion

After analysis using three methods (NMR, MS, IR), it can be concluded that the unknown chemical B is tert-Butyl Chloride (1-chloro-3-methylpropane). The chemical structure of unknown B is shown below.



References

|Keller L, Joseph K, and Miguel S. 2022 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (AL): Auburn University